ntRoot: Scalable Ancestry Predictions

from Genome Sequencing Data

BC CANCER RESEARCH
Provincial Health Services Authority
Canada's Michael Smith Genome Sciences Centre

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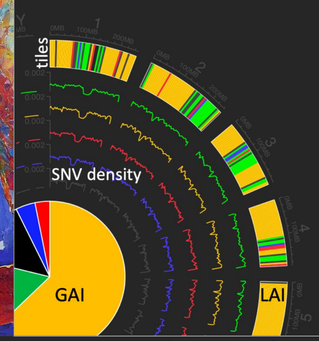
Ancestry Inference

- **Global (GAI)**
  Genome-wide summary
- **Local (LAI)**
  Estimation of ancestry along a DNA sequence



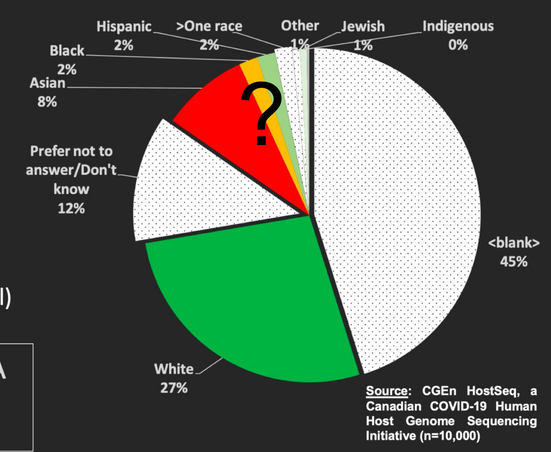
Continental/Geographic-level Super-Population (sp) Ancestry

- AMR Americas
- AFR Africa
- EUR Europe
- SAS South Asia
- EAS East Asia

Open Problems

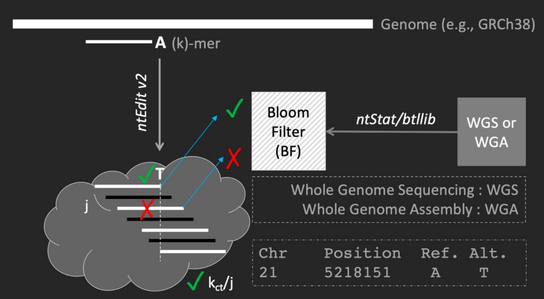
- Self-declared ancestry
  - Incomplete
  - Low resolution
  - Ambiguous
- Existing approaches
  - Do not scale well
  - Not streamlined
  - Limited input (e.g., WGS only)
  - Limited output (e.g., single GAI label)

Ancestry encoded in our DNA
Single Nucleotide Variants (SNVs)
Cohort studies include sequencing

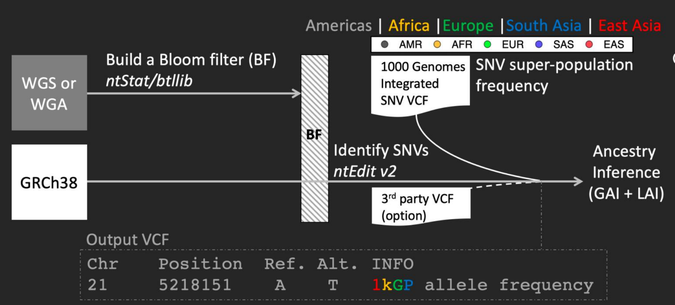


SNV Identification

sequence alignment-free



ntRoot Framework

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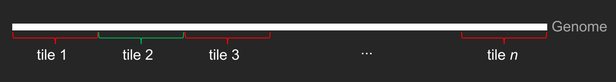
21  5218151  .  A  T  .  PASS  AD=19,19;EAS_AF=0.05;EUR_AF=0.05;AFR_AF=0.06;AMR_AF=0.04;SAS_AF=0.1;
```

GAI = score$_{sp}$

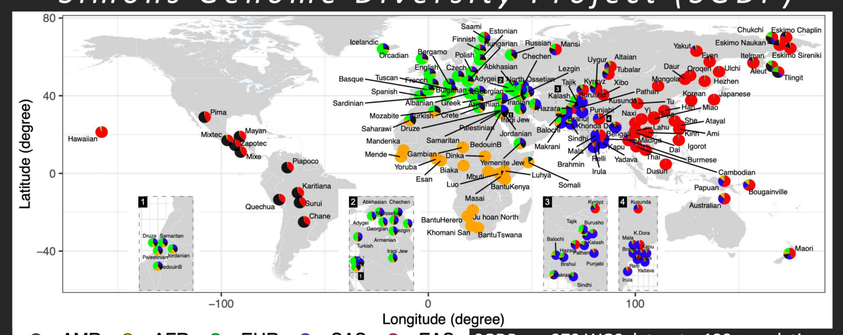
$$Average\ AF\ x\ nz\ SNV\ rate \qquad nz\ SNV\ rate_{sp} = \frac{nz\ AF\ SNV\ count_{sp}}{total\ nz\ AF\ SNV\ count}$$

$$LAI_fraction_{sp} = \frac{\sum_{i=1}^{num_tiles} length(tile_i)\ if\ tile_{i_{super-population}} == sp}{\sum_{i=1}^{num_tiles} length(tile_i)}$$

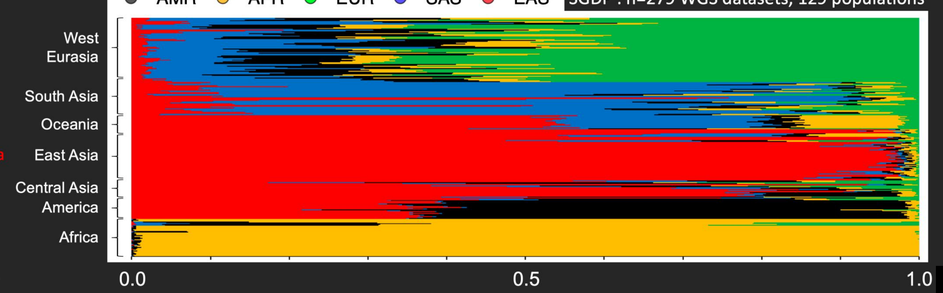


LAI-based Continental-level GAI Fractions

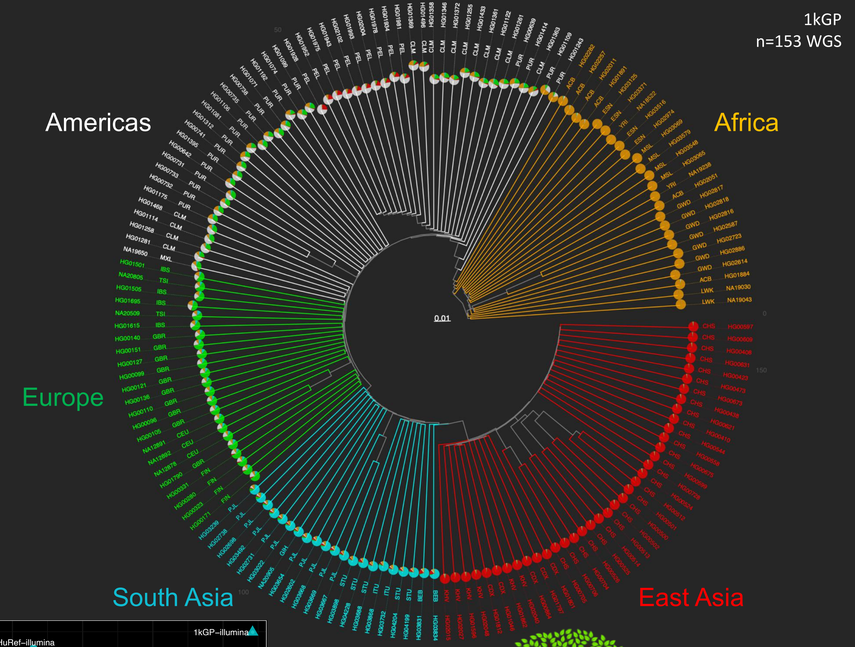
Simons Genome Diversity Project (SGDP)

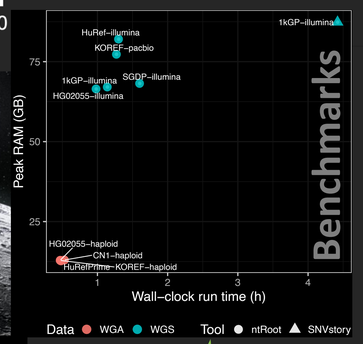


AMR AFR EUR SAS EAS | SGDP : n=279 WGS datasets, 129 populations



1kGP Phylogeny with Ancestry-Discriminant SNVs





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NTROOT

Scalable Continental-level Ancestry Predictions

Accurate • Streamlined • Fast • Low RAM

warrenlr.github.io | birollab.ca | github.com/bcgsc/ntroot